

GUANGBO YU

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EDUCATION

University of California, Irvine	Irvine, CA
Ph.D., Biomedical Engineering	Expected 2026
University of Southern California	Los Angeles, CA
M.S., Computer Science	2017
University of Electronic Science and Technology of China	Chengdu, China
B.S., Software Engineering	2015

CORE SKILLS

- **Machine Learning:** Recommendation Systems, Ranking & Personalization, Reinforcement Learning, Contextual Bandits, Embeddings & Representation Learning, Production ML
- **ML/DL Frameworks:** PyTorch (DDP), Hugging Face (TRL/PEFT), Scikit-learn, XGBoost, ONNX, FAISS
- **Architectures:** Transformers, VLMs, Two-Tower Retrieval, DIN/DCN-v2/MMoE, RQ-VAE, GRPO/RLVR, LoRA/QLoRA
- **Engineering & Tools:** Python, SQL, Docker, Git, CI/CD, FastAPI, ONNX Runtime, MLflow, Prometheus

WORK EXPERIENCE

University of California, Irvine	Irvine, CA
ML Research Engineer	2022 – Present
• Led ML research for a \$2M+ NIH-funded program spanning multimodal foundation models, representation learning, and outcome prediction; authored 21 peer-reviewed papers (10 first/co-first) and 8 under review. • Adapted frozen self-supervised vision foundation models (DINOv3, MedSAM) to volumetric imaging via spatial patch-token pooling + lightweight heads; beat end-to-end fine-tuning (AUROC 0.63) and a domain-specific 3D FM, reaching AUROC up to 0.96 with sub-second, under-100 KB deployed heads. • Post-trained vision-language foundation models with QLoRA SFT + GRPO/RLVR reinforcement learning; designed verifiable reward functions and a reproducible evaluation harness that improved reasoning accuracy and out-of-distribution generalization across 4 benchmarks. • Extended representation-learning methods to recommendation, ranking, and personalization: built a multimodal two-tower retrieval → ranking → serving system that beat ID-only baselines by 60%+, validated with offline A/B, off-policy, and counterfactual evaluation; served at sub-15 ms p99.	
Tencent QTrade	Shenzhen, China
NLP Engineer	2020 – 2022
• Replaced legacy LSTM with ALBERT-based NER via domain-specific continued pretraining, improving Macro-F1 by 11 points (77% → 88%); trained with PyTorch DDP across 8×V100 GPUs. • Architected and scaled an ML microservice (Flask/Gunicorn, NGINX) to 500k+ DAU with sub-100ms P95 latency; implemented canary rollout (5% → 100%) with online/offline monitoring and Jenkins-based model rollback. • Built a production data pipeline ingesting streaming data with rule-based preprocessing and domain corpus normalization; designed modular post-processing for output boundary repair, achieving 99%+ specificity. • Redesigned the system for extensibility by rewriting post-processing from scratch to support diverse data patterns and eliminate rule conflicts; led a team of 4 (2 MLEs, 2 annotators) through the full migration.	

Multimodal Recommendation & Ranking System with Semantic IDs

2025

Tech Stack: PyTorch, Qwen2.5-VL, FAISS/HNSW, RQ-VAE, ONNX, FastAPI, Docker, MLflow

- Built an **end-to-end multimodal recommender** (retrieval → ranking → serving) on **6.6M interactions**; **multimodal two-tower beat ID-only retrieval** by **+66–75% Recall@100** (+81% on cold-start), replicated across two product categories.
- Generated ranking features via **batched Qwen2.5-VL inference over 233K items**; added **RQ-VAE semantic IDs** for **+46–47% cold-start retrieval**, benchmarked CPU serving at **10–16 ms p99**, and fixed cascade bias for **up to +48% NDCG@10**.
- Turned cold-start into a live product feature: an id-free online path (content-pooled user embedding + behavior-sequence ranker) serves **session-based personalization for brand-new users with zero retraining**.

Post-Training Vision-Language Models with RLVR

2026

Tech Stack: PyTorch, Hugging Face TRL/PEFT, QLoRA, GRPO, vLLM, Weights & Biases

- Post-trained **Qwen3-VL (2B/8B)** for VQA on a single 24GB GPU with **QLoRA SFT + GRPO reinforcement learning from verifiable rewards**; engineered exact-match and format **reward functions** with vLLM-colocated rollouts, lifting reasoning-mode accuracy **+4.6 on VQAv2** and **+2.5 on held-out TextVQA (OOD)**.
- Built a contamination-free **four-benchmark evaluation harness** (image-disjoint splits, official VQAv2 metric) cross-checked against Imms-eval; a budget-capped **LLM-as-judge** found **23–49% of exact-match “misses” were false negatives**, correcting benchmark scores.
- Quantified a **“reasoning tax”** via a controlled **reward-weight ablation**: forcing chain-of-thought costs **3–13 points** on perception-saturated VQA, and GRPO rationally prunes reasoning when it lowers reward.
- Diagnosed a **TRL GRPO-VLM rollout bug** (dual image-preprocessing corrupting prompts) and contributed a fix **upstream (huggingface/trl#6401)**; released **4 post-training adapters on HuggingFace Hub** with a reproducible, seed-pinned repo.

Multilingual NER Framework

2022

ACL 2022, SemEval-2022 Task 11 (Top-3 in Multilingual)

- Built a semi-supervised NER pipeline: continued pretraining of **XLM-RoBERTa** on **20GB external corpus**, adversarial training (PGD), pseudo-label generation, and a **7-model ensemble** via majority voting.
- Placed **Top-3 in Multilingual** (Macro-F1: **84.35**), **Top-4 in Code-Mixed**, and **Top-7 in Chinese** tracks worldwide.

SELECTED PUBLICATIONS (5 OF 21 PEER-REVIEWED; 8 ADDITIONAL UNDER REVIEW)

- Yu, G. *et al.* Qtrade AI at SemEval-2022 Task 11: An Unified Framework for **Multilingual NER** Task. *ACL*, 2022.
- Yu, G. *et al.* **Multi-Task Deep Learning** on MRI for Tumor Segmentation and Treatment Response Prediction in an Experimental Model of Hepatocellular Carcinoma. *Diagnostics*, 2025.
- Yu, G. *et al.* **MRI Radiomics** to Monitor Therapeutic Outcome of Sorafenib plus IHA Transcatheter NK Cell Combination Therapy in Hepatocellular Carcinoma. *J. Transl. Med.*, 2024.
- Yu, G. *et al.* **Deep Learning** Prediction of MRI Data for Evaluating Combined Transcatheter NK Cell Local Delivery and Sorafenib Therapy in Hepatocellular Carcinoma. *JVIR*, 2025.
- Yu, G. *et al.* AI-Derived MRI Biomarkers Using **Vision Transformer** for Predicting Combination Immunotherapy Outcomes in Liver Cancer. *ISMRM*, 2025.